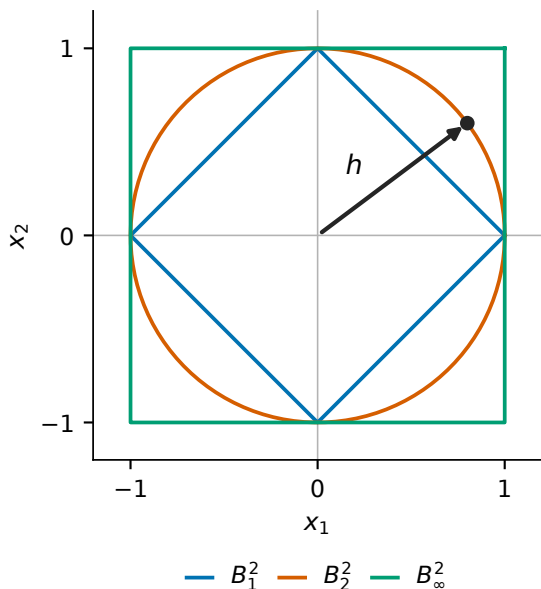
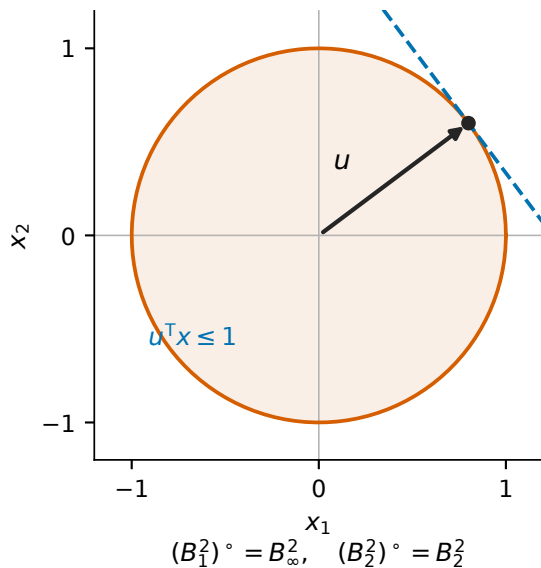


(a) Unit balls and distance



(b) Duality; self-duality at 2



$$h = u = (4/5, 3/5): \quad \|h\|_1 = 7/5, \quad \|h\|_2 = 1, \quad \|h\|_\infty = 4/5$$

$$p^{-1} + q^{-1} = 1: \quad \|y\|_q = \sup_{\|x\|_p \leq 1} |y^T x|$$

$$f = \sum_j f_j e_j \leftrightarrow (f_j) \in \ell^2, \quad \|f - g\|_{\ell^2}^2 = \sum_j |f_j - g_j|^2$$

$$\Lambda_\beta(f) = \langle f, \beta \rangle_{L^2}$$